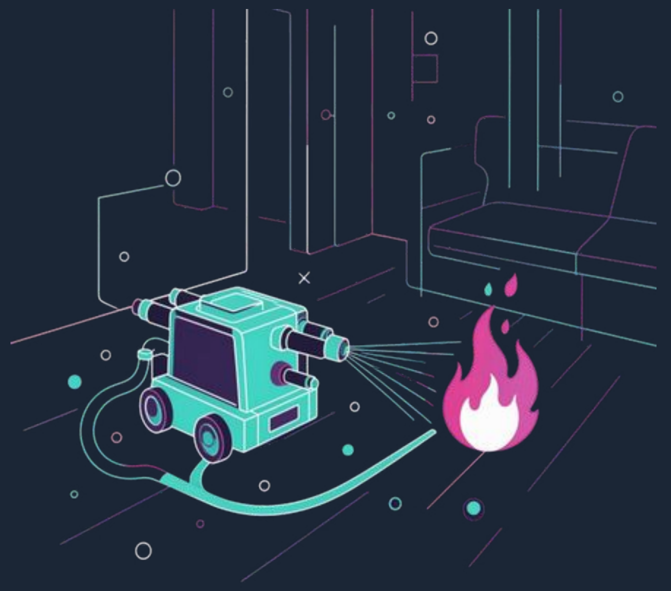
# Fire Fighting Bot

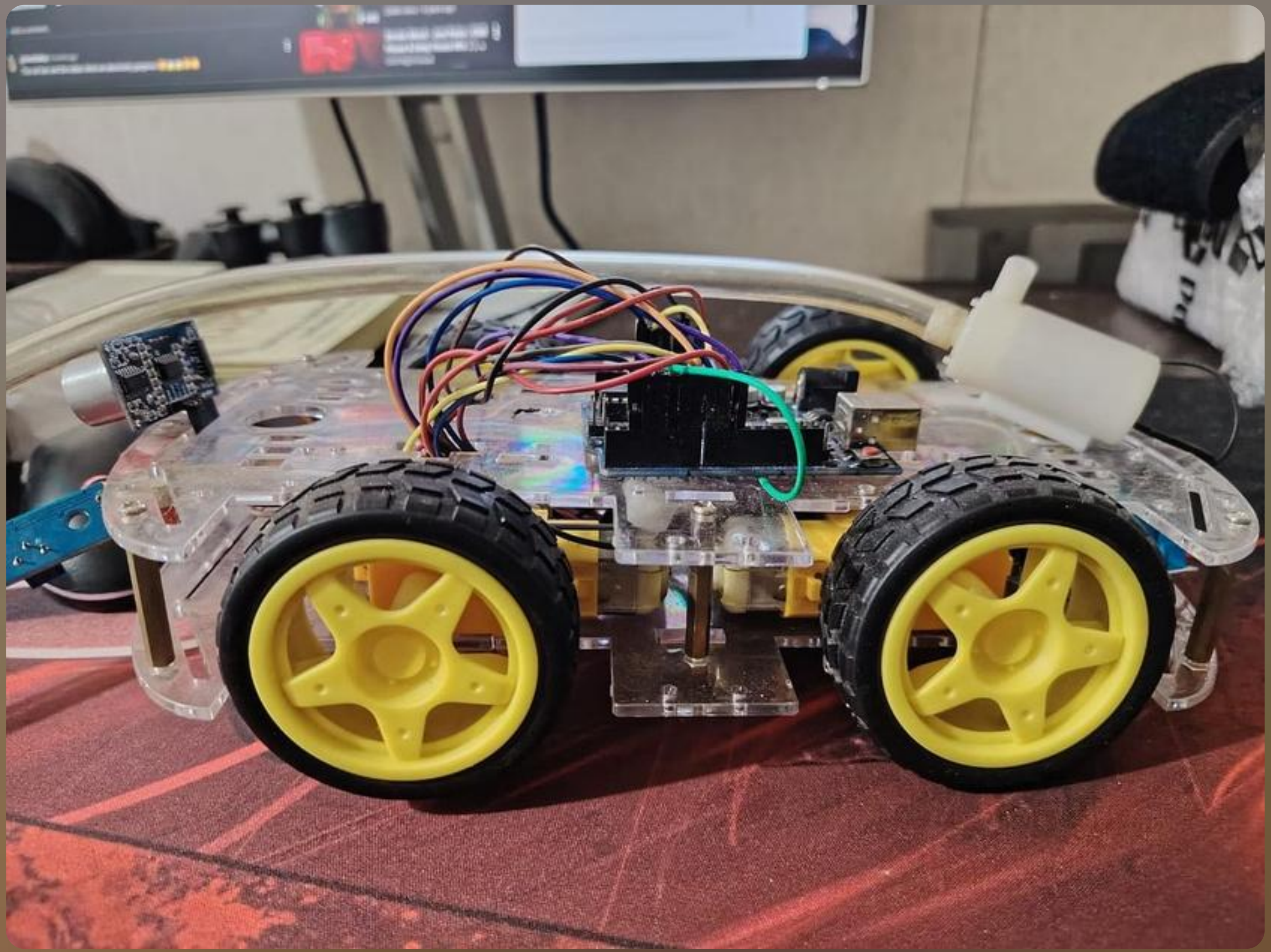
Arduino-based mobile robot designed to detect small fires and extinguish them remotely, reducing human risk.

# Purpose



Designed to reduce human risk during small fire incidents by autonomously detecting fires and spraying water to extinguish them.

- Autonomous detection
- Navigation toward fire
- Targeted water spraying



# Working Process

Detects flame → movestowardsource → aimsnozzle with servo → activates pumpto spray water.

# Key Hardware Components

Arduino Uno

Motor Driver

Servo Motor

Relay Module

Robot Chassis

USB Cable

Capacitor

Flame Sensor

Ultrasonic Sensor

# Fire Detection & Extinguishing Hardware

### Flame Sensors

Multiple modules positioned around the chassis to detect flame direction (VCC, GND, analog output).

### Water Pump & Tank

Pump activated via relay to push water through nozzle when fire is detected.

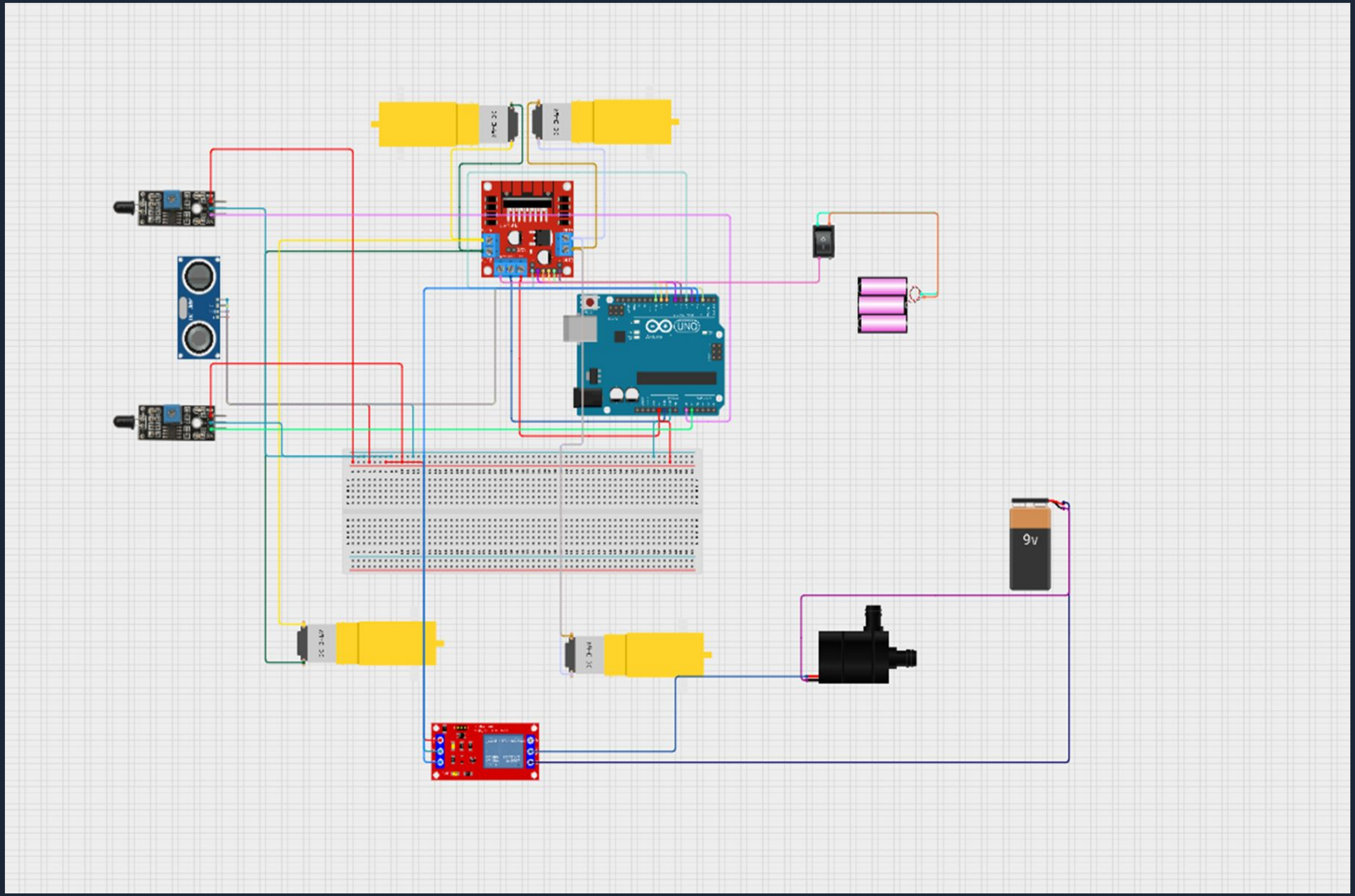
### Nozzle & Aim Servo

Servo adjusts nozzle direction for targeted extinguishing.

### Relay & Power

Relay modules tested for response; 9V battery or external supply powers the system.

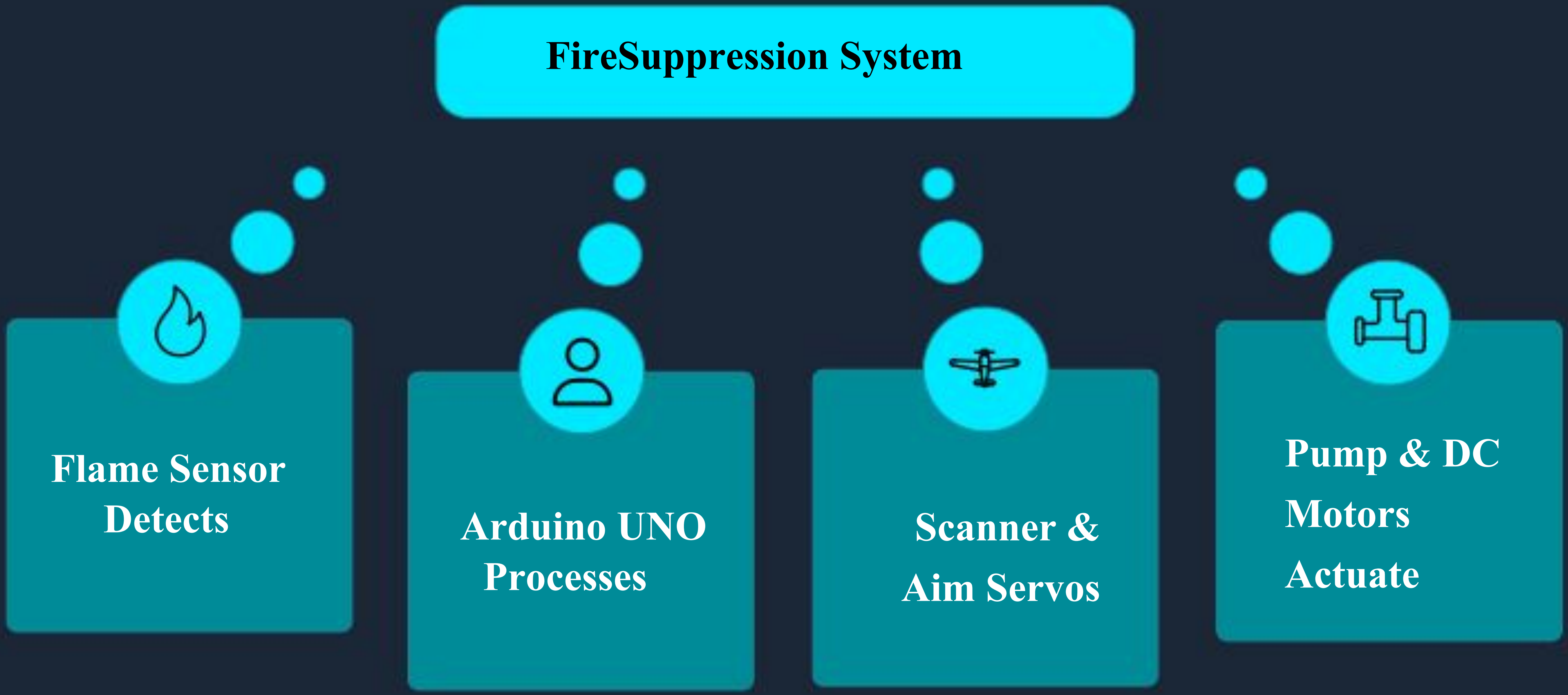
# Circuit Diagram Overview




The Arduino Uno connects to four flame sensors, servos (scanner and aim), an L298N motor driver for DC motors, a relay controlling the water pump, a buzzer, and a 9V power source.

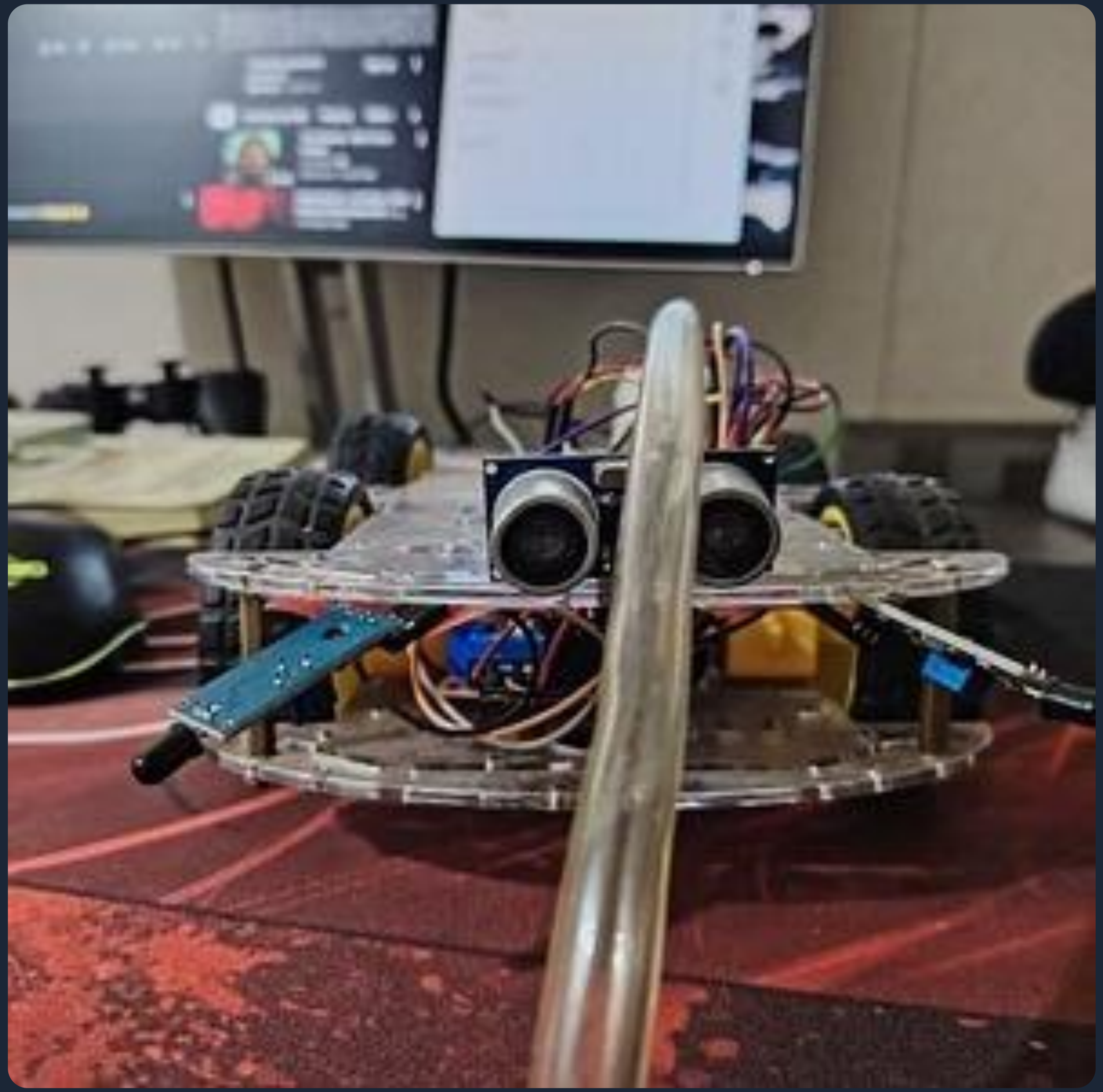
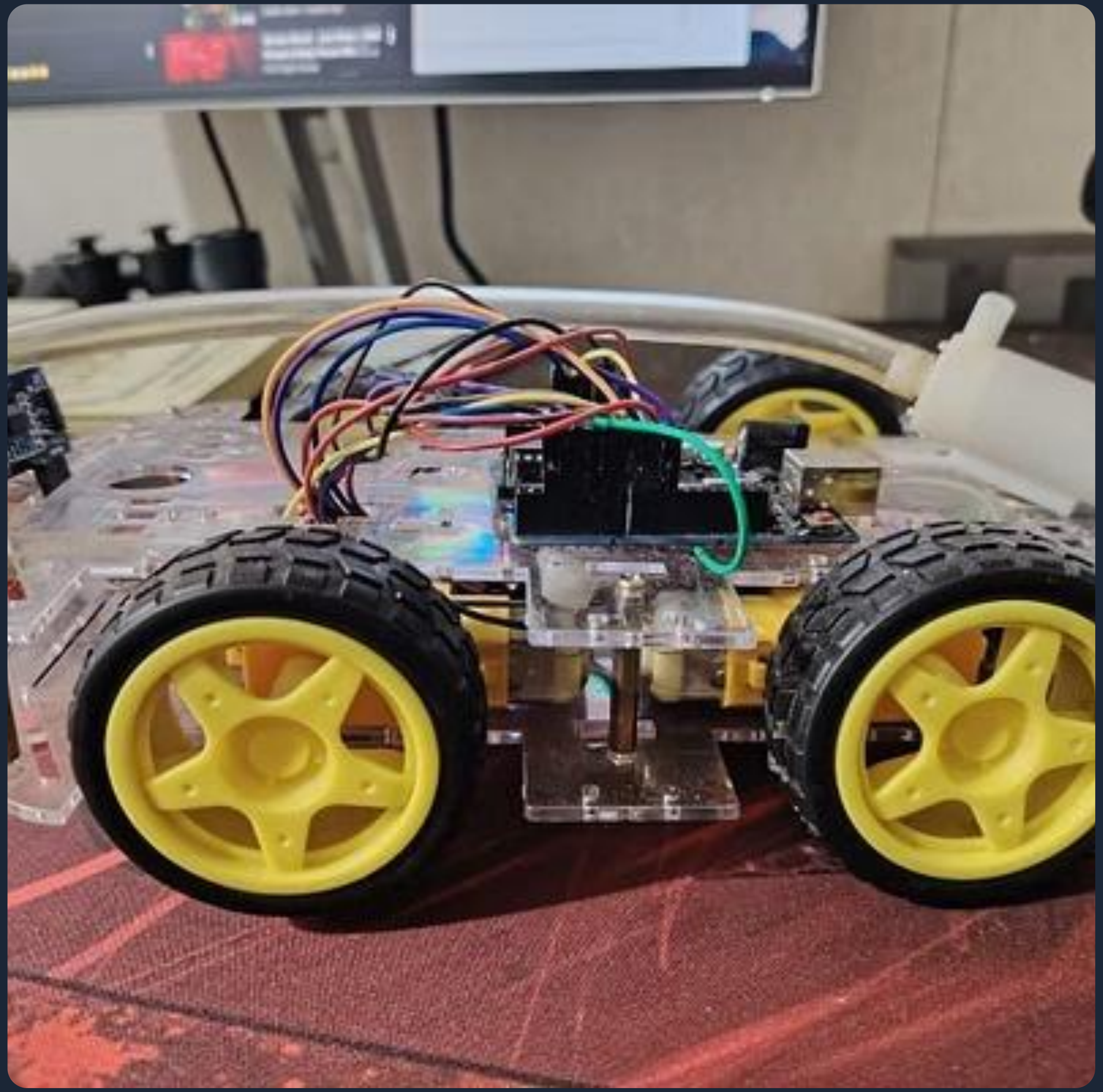
Visual anchor: central Arduino coordinating sensor inputs and actuator outputs.

# Embedded System




This flowchart shows flame detection feeding the Arduino, which commands scanning, nozzle aiming, drive motors, and pump activation.

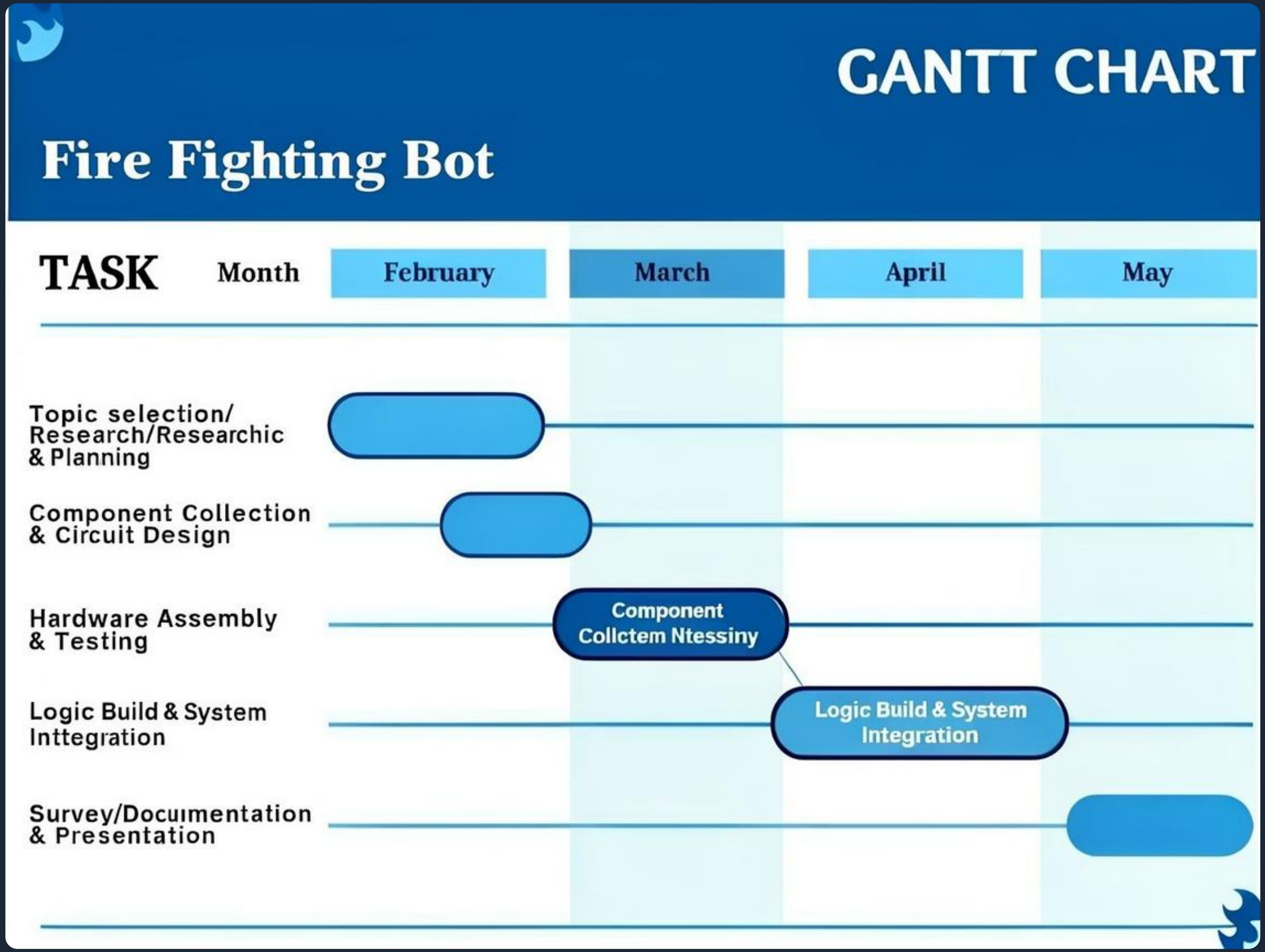
# Assembly Photos & Video

Photos show chassis, breadboard wiring, flame sensor close-up, and assembled prototype.


Obstacles Avoiding with firefighter bot

# Timeline (Gantt Chart)




Project phases :

- Topic selection, Research & Planning 4 Feb
- Component Collection & Circuit Design

  4 Mar
- Hardware Assembly & Testing 4 Mar
-
- Logic Build & System Integration 4 Apr
- Survey, Documentation & Presentation

  4 May

# System Logic & Flow Diagram

# Conclusion & Thanks

FlameGuard demonstrates a compact, Arduino-driven approach to reducing human risk during small fires by combining multi-sensor detection, targeted nozzle aiming, and pump-controlled extinguishing.

**Thank You**

Any questions?